

TUGAS ASYNCHRONOUS (VIEW)

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1. Buatlah tiga tabel sistem perpustakaan yang terdiri dari tabel book, user, dan flow dengan kolom:
 - a. Book: *bookID*, *bookTitle*, *authorName*, *borrowedStatus*

```
MariaDB [classicmodels]> DESCRIBE books;
```

Field	Type	Null	Key	Default	Extra
bookID	int(11)	NO	PRI	NULL	auto_increment
bookTitle	varchar(255)	NO		NULL	
authorName	varchar(255)	NO		NULL	
borrowedStatus	enum('borrowed','available')	NO		NULL	

4 rows in set (0.041 sec)

```
MariaDB [classicmodels]> SELECT * FROM books;
```

bookID	bookTitle	authorName	borrowedStatus
100	Matematika Diskrit	Scarlett	borrowed
101	Gizi Dalam Daur Kehidupan	Yulia	available
102	Basis Data	Thomas	borrowed
103	Gizi Olah Raga	Naufal	borrowed
104	Algoritma	Einstein	borrowed
105	Pengantar Ekonomi Mikro	Mubarak	available
106	Ekonomi Moneter	Aksata	borrowed
107	Harry Potter	J.K Rowling	borrowed
108	Educated	Tara	borrowed
109	Statistik untuk Pemula	Budi	available
110	Belajar SQL dengan Mudah	Dimas	borrowed
111	Logika Matematika Dasar	Siti	borrowed
112	Manajemen Waktu Efektif	Teguh	available
113	Etika Profesi Dietisien	Ahmad	borrowed
114	Pengenalan Data dan Informasi	Laila	available
115	Biologi	Alya	borrowed
116	Kimia Organik	Hather	borrowed
117	Anatomi Manusia	Putri	borrowed
118	Music	Charlie	borrowed
119	Dietetik	Elin	borrowed
120	Kopi	Dewi	borrowed
121	Design Interior	Utami	borrowed
122	Pengantar Filsafat Hukum	Diva	borrowed

- b. Flow: *flowID*, *userIDBorrowing*, *bookIDBorrowed*, *borrowDate*, *returnDate*

```
MariaDB [classicmodels]> DESCRIBE flow;
```

Field	Type	Null	Key	Default	Extra
flowID	int(11)	NO	PRI	NULL	auto_increment
userIDBorrowing	int(11)	NO		NULL	
bookIDBorrowed	int(11)	NO		NULL	
borrowdate	date	NO		NULL	
returndate	date	NO		NULL	

5 rows in set (0.039 sec)

```
MariaDB [classicmodels]> SELECT * FROM flow;
```

flowID	userIDBorrowing	bookIDBorrowed	borrowdate	returndate
1	1000	100	2025-01-27	2025-02-10
2	1002	103	2025-02-10	2025-02-17
3	1003	104	2025-02-20	2025-02-25
4	1004	106	2025-02-21	2025-02-27
5	1005	108	2025-03-15	2025-03-20
6	1006	110	2025-03-17	2025-03-28
7	1007	111	2025-03-28	2025-04-01
8	1006	113	2025-05-12	0000-00-00
9	1008	102	2025-05-12	0000-00-00
10	1009	107	2025-05-12	0000-00-00
11	1004	116	2025-05-10	0000-00-00
12	1004	118	2025-05-10	0000-00-00
13	1004	119	2025-05-09	0000-00-00
14	1006	115	2025-05-06	0000-00-00
15	1006	117	2025-05-07	0000-00-00
16	1007	120	2025-05-11	0000-00-00
17	1007	121	2025-05-11	0000-00-00
18	1007	122	2025-05-11	0000-00-00

- c. User: *userID*, *userName*, *numberOfBorrowing*, *numberOfReturning*

```
MariaDB [classicmodels]> DESCRIBE user;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| userID | int(11) | NO | PRI | NULL | auto_increment |
| userName | varchar(255) | NO | | NULL | |
| numberOfborrowing | int(11) | YES | | NULL | |
| numberOfreturning | int(11) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.039 sec)

MariaDB [classicmodels]> SELECT * FROM user;
+-----+-----+-----+-----+
| userID | userName | numberOfborrowing | numberOfreturning |
+-----+-----+-----+-----+
| 1000 | andiwira | 1 | 1 |
| 1001 | lilira | NULL | NULL |
| 1002 | agusmora | 2 | 1 |
| 1003 | rinadewi | 2 | 2 |
| 1004 | bimasena | 4 | 1 |
| 1005 | sitimaya | 1 | 1 |
| 1006 | bayuperkasa | 4 | 1 |
| 1007 | fitrianur | 4 | 1 |
| 1008 | teguhaji | 1 | NULL |
| 1009 | melatisari | 1 | NULL |
+-----+-----+-----+-----+
```

2. Gunakan VIEW untuk menyelesaikan persoalan berikut:
- Buat VIEW untuk menampilkan semua judul buku yang dipinjam kemarin, lengkap dengan status dan tanggal peminjaman.

```
MariaDB [classicmodels]> CREATE VIEW books_update AS
-> SELECT b.bookTitle, b.borrowedStatus, f.borrowdate
-> FROM books as b
-> JOIN flow as f ON b.bookID = f.bookIDBorrowed
-> WHERE f.borrowdate = CURDATE() - INTERVAL 1 DAY;
Query OK, 0 rows affected (0.012 sec)

MariaDB [classicmodels]> SELECT * FROM books_update;
+-----+-----+-----+
| bookTitle          | borrowedStatus | borrowdate |
+-----+-----+-----+
| Etika Profesi Dietisien | borrowed      | 2025-05-12 |
| Basis Data         | borrowed      | 2025-05-12 |
| Harry Potter       | borrowed      | 2025-05-12 |
+-----+-----+-----+
3 rows in set (0.002 sec)
```

- Buat VIEW untuk menampilkan semua judul buku, baik yang dipinjam maupun tidak dipinjam, serta userID peminjam untuk buku yang sedang dipinjam.

```
MariaDB [classicmodels]> CREATE VIEW book_status AS
-> SELECT b.bookTitle, f.userIDBorrowing
-> FROM books as b
-> LEFT JOIN flow as f ON b.bookid = f.bookIDBorrowed
-> ;
Query OK, 0 rows affected (0.013 sec)

MariaDB [classicmodels]> SELECT * FROM book_status;
+-----+-----+
| bookTitle          | userIDBorrowing |
+-----+-----+
| Matematika Diskrit | 1000            |
| Gizi Olah Raga     | 1002            |
| Algoritma          | 1003            |
| Ekonomi Moneter    | 1004            |
| Educated           | 1005            |
| Belajar SQL dengan Mudah | 1006          |
| Logika Matematika Dasar | 1007          |
| Etika Profesi Dietisien | 1006          |
| Basis Data         | 1008            |
| Harry Potter       | 1009            |
| Gizi Dalam Daur Kehidupan | NULL          |
| Pengantar Ekonomi Mikro | NULL          |
| Statistik untuk Pemula | NULL          |
| Manajemen Waktu Efektif | NULL          |
| Pengenalan Data dan Informasi | NULL          |
+-----+-----+
15 rows in set (0.001 sec)
```

- c. Buat VIEW untuk menampilkan daftar buku yang dipinjam beserta semua userID yang terlibat dalam aktivitas peminjaman atau pengembalian.

```
MariaDB [classicmodels]> CREATE VIEW book_activity AS
-> SELECT b.booktitle, f.userIDBorrowing
-> FROM books as b
-> JOIN flow as f ON b.bookID = f.bookIDBorrowed;
Query OK, 0 rows affected (0.012 sec)

MariaDB [classicmodels]> SELECT * FROM book_activity;
+-----+-----+
| booktitle          | userIDBorrowing |
+-----+-----+
| Matematika Diskrit | 1000            |
| Gizi Olah Raga     | 1002            |
| Algoritma          | 1003            |
| Ekonomi Moneter    | 1004            |
| Educated           | 1005            |
| Belajar SQL dengan Mudah | 1006          |
| Logika Matematika Dasar | 1007          |
| Etika Profesi Dietisien | 1006          |
| Basis Data         | 1008            |
| Harry Potter       | 1009            |
+-----+-----+
10 rows in set (0.001 sec)
```

- d. Buat VIEW yang menampilkan daftar judul buku dan userName yang meminjam buku serta hanya menampilkan user yang meminjam lebih dari 3 buku.

```
MariaDB [classicmodels]> CREATE VIEW frequent_users AS
-> SELECT b.booktitle, u.username
-> FROM flow as f
-> JOIN books as b ON f.bookIDBorrowed = b.bookID
-> JOIN user as u ON f.userIDBorrowing = u.userID
-> WHERE f.userIDBorrowing IN (
->   SELECT userIDBorrowing
->   FROM flow
->   GROUP BY userIDBorrowing
->   HAVING COUNT(*) > 3
-> );
Query OK, 0 rows affected (0.014 sec)

MariaDB [classicmodels]> SELECT * FROM frequent_users;
+-----+-----+
| booktitle          | username        |
+-----+-----+
| Ekonomi Moneter    | bimasena        |
| Belajar SQL dengan Mudah | bayuperkasa    |
| Logika Matematika Dasar | fitrianur       |
| Etika Profesi Dietisien | bayuperkasa     |
| Kimia Organik      | bimasena        |
| Music              | bimasena        |
| Dietetik           | bimasena        |
| Biologi            | bayuperkasa     |
| Anatomi Manusia    | bayuperkasa     |
| Kopi               | fitrianur       |
| Design Interior    | fitrianur       |
| Pengantar Filsafat Hukum | fitrianur       |
+-----+-----+
12 rows in set (0.003 sec)
```